

BILL HUYNH

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EDUCATION

California State University, Sacramento

Bachelor of Science in Computer Engineering | GPA: 3.55

Sacramento, CA

May 2024

EXPERIENCE

Self-Employed Full Stack Engineer

Independent

May 2024 – Present

Milpitas, CA

- Built and shipped **4** production applications end-to-end, including AI pipelines, event-driven backends, and cloud infrastructure serving **200,000+** monthly requests.
- Architected full-stack systems across frontend interfaces, backend APIs, storage workflows, asynchronous job pipelines, and cloud deployment using TypeScript, Next.js, Express, Python, FastAPI, Supabase, and AWS.
- Improved reliability for long-running AI and media-processing workloads through event-driven orchestration, retry handling, reconciliation, and serverless automation.

Undergraduate Researcher

Texas A&M University Kingsville

May 2023 – Aug 2023

Kingsville, TX

- Awarded **Best Paper** at **IEEE PICOM 2025** for wireless sensor network optimization research.
- Improved wireless sensor network coverage efficiency by **40%** by developing an optimized algorithm.
- Validated findings through **1,000+** simulations with faculty and peer researchers.

PROJECTS

Disburse: AI Video Repurposing Platform | Next.js, TypeScript, Python, FastAPI, FFmpeg, PostgreSQL, Supabase, AWS

- Designed a distributed workflow coordinating **6** asynchronous processing stages, including transcript ingestion, clip generation, facecam analysis, edit configuration, and FFmpeg-based rendering.
- Developed facecam-aware vertical layout generation capable of sampling **400+** frames per detection run to automatically format videos for TikTok and YouTube Shorts.
- Reduced pipeline execution failures for **60+** minute AI/media-processing workloads by implementing event-driven orchestration, retry handling, and reconciliation systems.

Yaps: AI Powered Learning Platform | Next.js, TypeScript, Express, Fly.io, Stripe, AWS S3

- Deployed a full-stack SaaS that converts uploaded MP4/MP3 lectures into AI-narrated podcast lessons with auto-generated summaries and flashcards.
- Migrated backend to event-driven architecture using **Inngest**, eliminating request timeouts and enabling processing of **60+** minute videos by decoupling long-running AI and ffmpeg jobs from the web API.
- Reduced LLM summarization cost by **45%** via semantic chunking and prompt tuning while preserving output quality for TTS and flashcard generation.

TFT-dle | TypeScript, React, Python, Selenium, AWS Lambda/CloudFront, Supabase

- Deployed a character guessing game that supports **100+** daily active users and **200,000+** monthly requests.
- Optimized frontend state and backend APIs to achieve **<1s** page load times and **99.9%** uptime.
- Automated a serverless scraping pipeline using **Python**, **Selenium**, and **AWS Lambda** to process **2,000+** images and **150+** videos from live TFT datasets.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, C++, HTML, CSS

Frontend: React, Next.js, Tailwind CSS

Backend & APIs: Node.js, Express, FastAPI, Spring Boot, REST APIs, Drizzle ORM, PostgreSQL

Cloud & Infrastructure: AWS (EC2, Lambda, S3, API Gateway, CloudFront), Supabase, Fly.io, Vercel, Docker